Introduction to R and the tidyverse

Exercise 1 — ISS dataviz, Spring 2026

YOUR NAME HERE

Invalid Date

Table of contents

Task 1: Session check-in

Three interesting or exciting things:

1. Something
2. Something
3. Something

Three muddy or unclear things:

1. Something
2. Something
3. Something

Task 2: Penguins

Data description

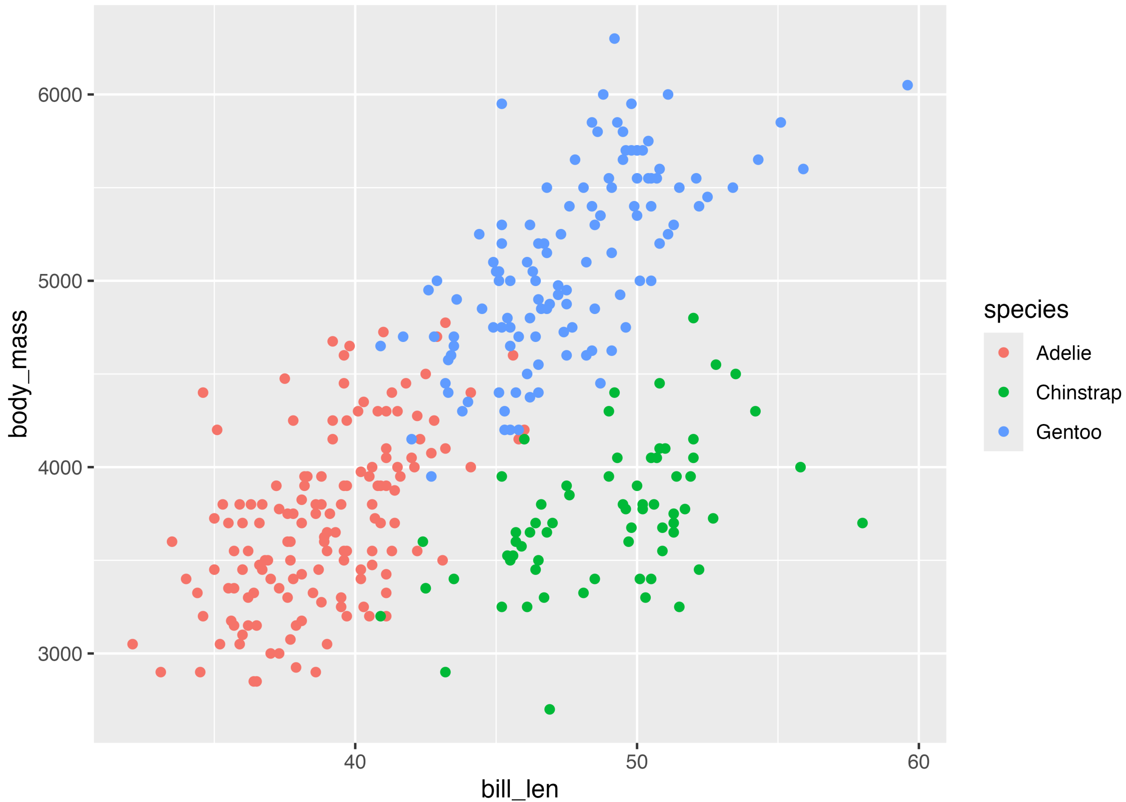
For this exercise, you'll be exploring data about 333 penguins living across three islands in Antarctica. See this website for complete details about the data and the variables included.

```
library(tidyverse)
```

```r
penguins <- read_csv("data/penguins.csv") |>
  drop_na(sex)
```

Recreate scatterplot

Run the code chunk below to see a plot.

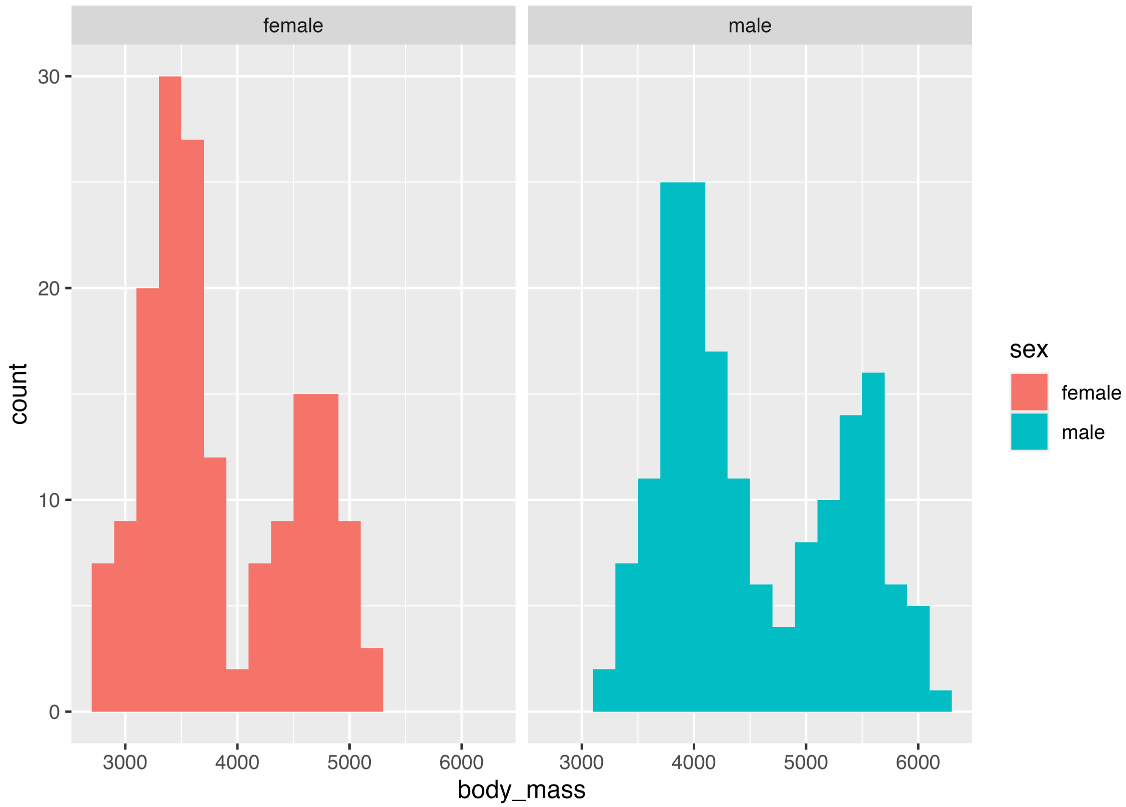


Use `ggplot()` in the chunk below to re-create the plot above.

```r
# Do stuff here
```

Recreate histogram

Run the code chunk below to see a plot.



Use `ggplot()` in the chunk below to re-create the plot above (hint—it's a histogram with a bin width of 200 grams).

```
# Do stuff here
```

Recreate scatterplot

Run the code chunk below to see a plot.

Insert a new chunk below and use `ggplot()` re-create the plot above.

Recreate filtered dataset

Here are all the Chinstrap penguins that weigh more than 4,000 grams:

```
# A tibble: 15 × 8
   species   island bill_len bill_dep flipper_len body_mass sex     year
   <chr>     <chr>     <dbl>    <dbl>       <dbl>     <dbl> <chr>  <dbl>
 1 Chinstrap Dream      46       18.9         195      4150 female  2007
 2 Chinstrap Dream      52       18.1         201      4050 male    2007
 3 Chinstrap Dream      50.5     19.6         201      4050 male    2007
 4 Chinstrap Dream      49.2     18.2         195      4400 male    2007
 5 Chinstrap Dream      52       19           197      4150 male    2007
 6 Chinstrap Dream      52.8     20           205      4550 male    2008
 7 Chinstrap Dream      54.2     20.8         201      4300 male    2008
 8 Chinstrap Dream      51       18.8         203      4100 male    2008
 9 Chinstrap Dream      52       20.7         210      4800 male    2008
10 Chinstrap Dream      53.5     19.9         205      4500 male    2008
11 Chinstrap Dream      50.8     18.5         201      4450 male    2009
12 Chinstrap Dream      49       19.6         212      4300 male    2009
13 Chinstrap Dream      50.7     19.7         203      4050 male    2009
```

```
14 Chinstrap Dream       49.3      19.9          203      4050 male    2009
15 Chinstrap Dream       50.8      19            210      4100 male    2009
```

Use `filter()` in the chunk below to re-create that dataset:

```
# Do stuff here
```

Recreate summarized dataset

Here are the average weights of male penguins across the three islands:

```
# A tibble: 3 × 2
  island    avg_weight
  <chr>          <dbl>
1 Biscoe         5105.
2 Dream          3987.
3 Torgersen      4035.
```

Insert a chunk below and use `filter()`, `group_by()`, and `summarize()` to re-create that dataset:

Task 3: Extension

Copy the code for one of your plots or tables above and paste it into the chunk below. Do some extra things to it, like changing the labels, changing the colors, adding a new geom, plotting a different variable from the dataset, changing the theme, or whatever. **This is your chance to play around and explore.**

```
# Do stuff here
```

Bibliography